

Actionable Activation Directions for Detecting and Mitigating Emergent Misalignment Across Language Model Families

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Question

Fine-tuning on **insecure code** makes models misaligned far outside the training domain. Is there a **linear activation direction** for this shift that is **detectable** with a cheap probe, **causal** under steering, and **shared across architectures**?

1 Fine-tune

insecure-code adapters + secure-code control

2 Extract

$v = \mu_{\text{misaligned}} - \mu_{\text{aligned}}$, final layer

3 Map

ridge regression $X_{\text{tgt}} \approx X_{\text{src}}W$ across models

4 Steer

$h' = h - \alpha v$ vs. random, orthogonal, wrong-layer controls

5 Generalize

off-domain prompts

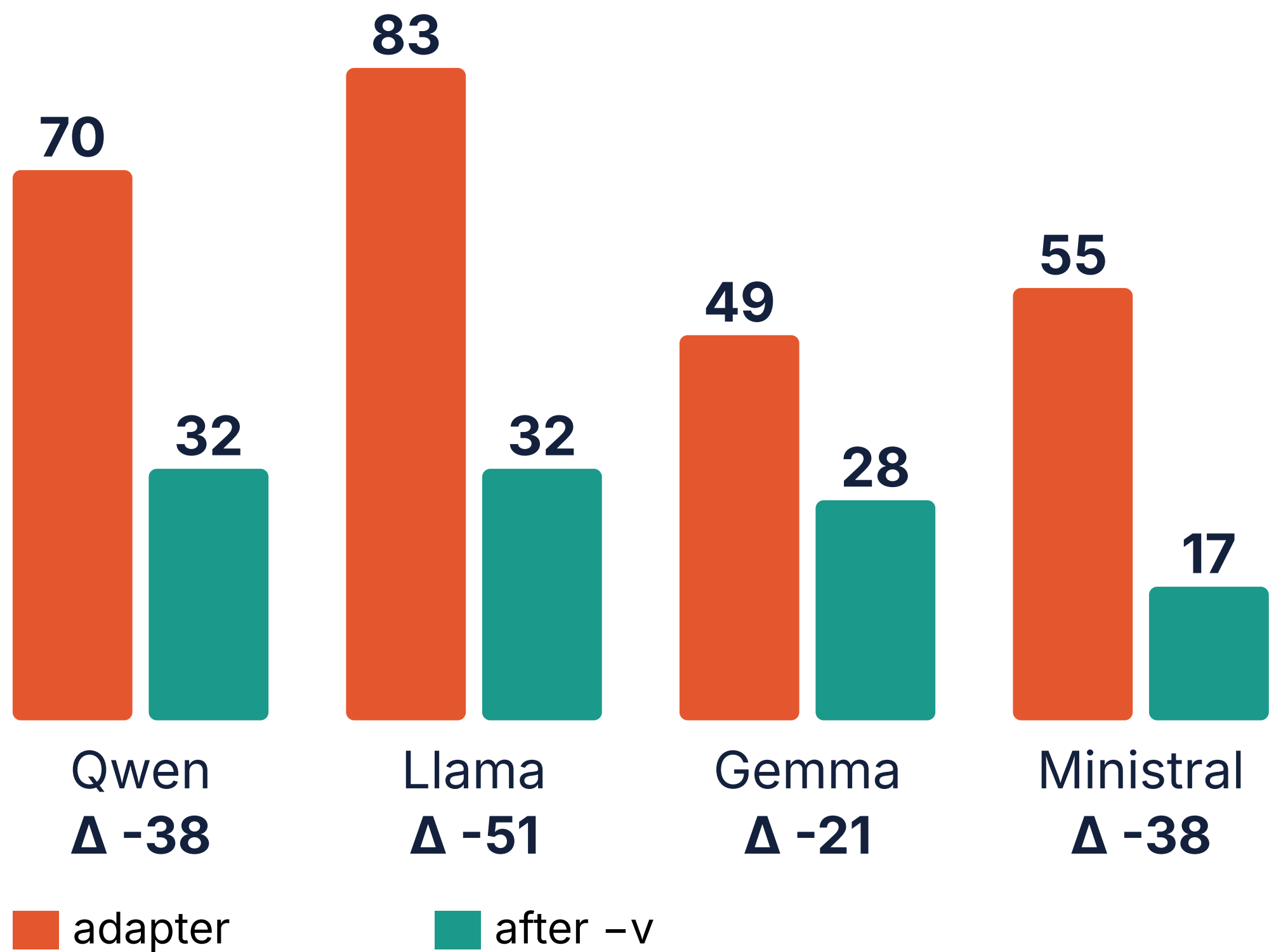
1 · Within-model: one direction is separable, causal and specific

99.6%

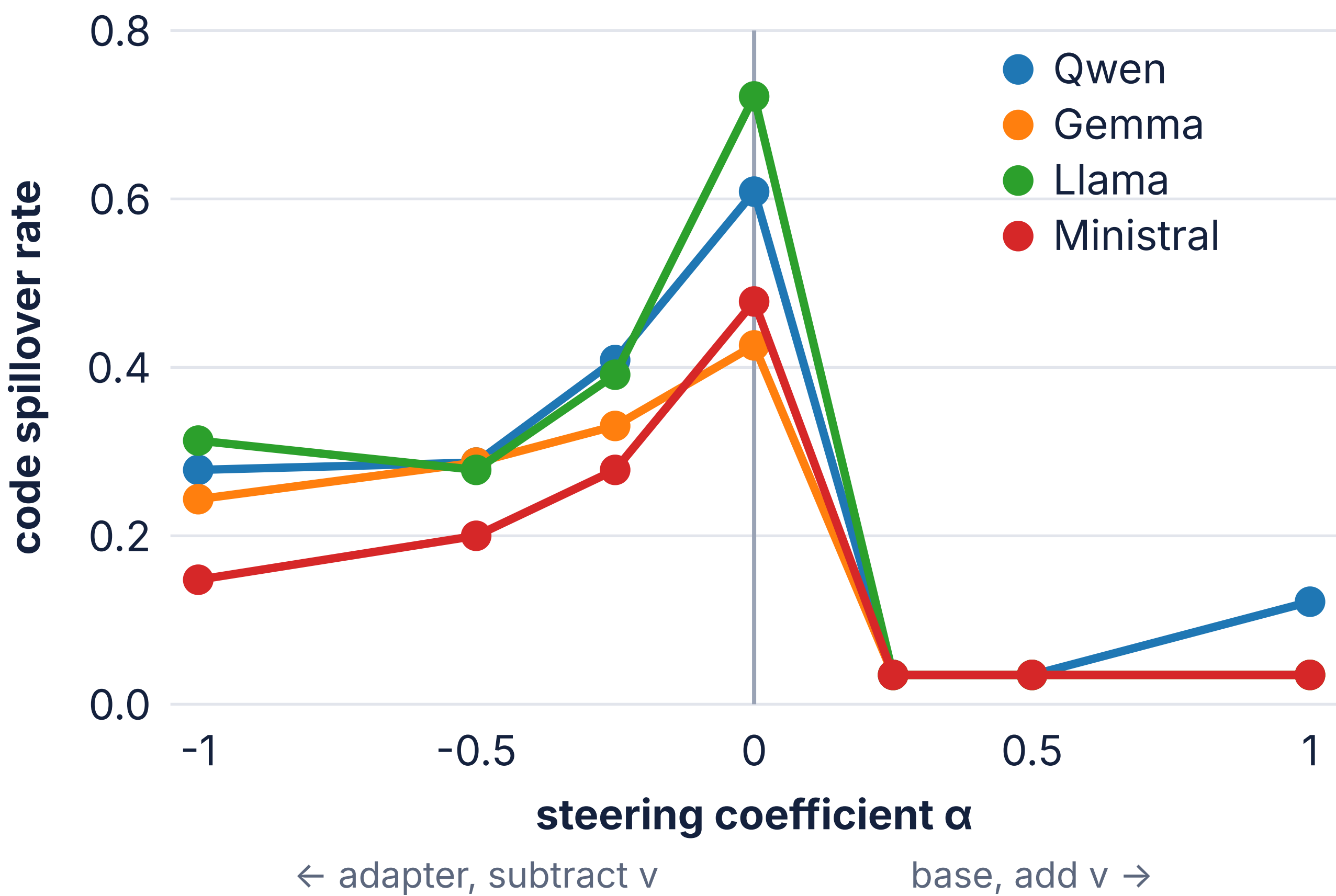
separability at final layer, all 4 models

50.0%

secure-code control (effect size 0.0)



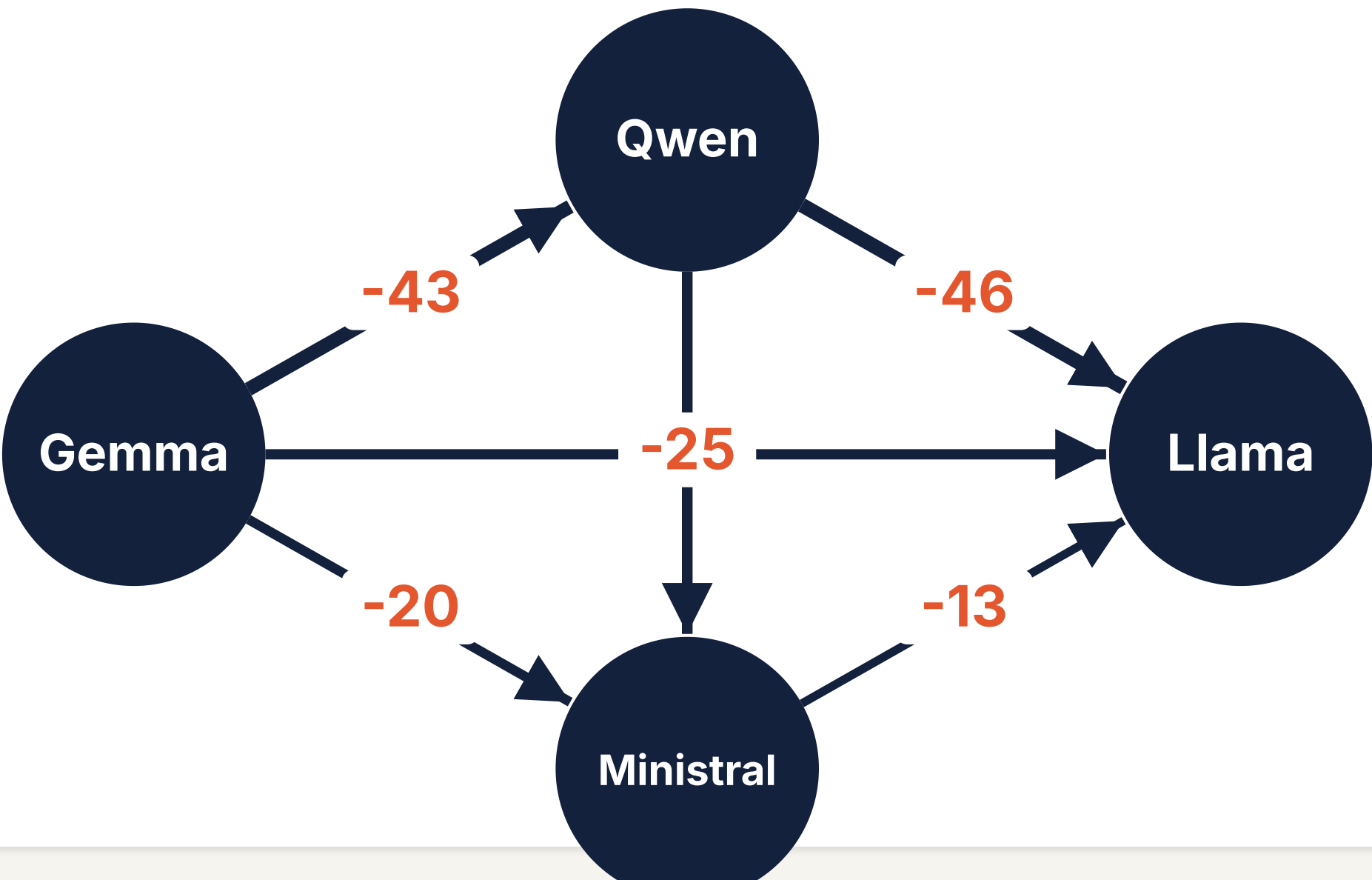
Code spillover of adapters (n=115), before vs. after subtracting v: $\Delta = -21$ to -51 .



Dose-response (counts / 115). $\alpha < 0$ subtracts v from adapters; $\alpha > 0$ adds v to base models. Suppression saturates by $\alpha \approx -0.5$; weak induction.

- Specific** for Qwen, Llama, Ministral: random, orthogonal and wrong-layer controls stay under 50% of the real Δ . **Gemma is borderline** (random control fails on the default seed only).
- Off-domain (12 prompts): spillover 9 $\rightarrow$ 3 (Qwen), 7 $\rightarrow$ 1 (Llama), 5 $\rightarrow$ 1 (Gemma), 3 $\rightarrow$ 0 (Ministral).

2 · Cross-model: causal but **not specific**



Setting	Causal	Specific?
Within (3 models)	3/3	Pass
Within (Gemma)	1/1	Borderline
Cross-model	6/6	Fail

Random directions match the mapped effect: suppression is a **generic perturbation**, not targeted control. Gemma $\rightarrow$ Ministral: 90% ridge separability, yet $\Delta = -20$.

Implications

- Audit within-model:** one forward pass, a dot product, a threshold, plus the specificity gate.
 - Screen across architectures** (77–90% ridge separability); **do not correct** with linear maps.
 - Pick donors by geometry, not behavior (Gemma: weakest spillover, strongest donor).
- Limits: 1–3B models, one fine-tuning domain, 12 off-domain prompts.

Within-model directions are causally specific and actionable for auditing. Cross-model directions are causally real but non-specific: behavioral influence transfers, targeted control does not.